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PI Name:	Holland, Murray John Ph.D.		
Project Title:	Generation of Fragile and Strongly Correlated Quantum States in Bose-Einstein Condensed Gases		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
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In this research program we investigated the phenomena of strongly correlated quantum states in quantum degenerate alkali gases. Strong interactions between the atoms arise from the presence of Feshbach scattering resonances which is the focus of a significant experimental effort at this time. Experiments in quantum degenerate gases have helped to elucidate many phenomena in superconductors and superfluid Helium systems by providing a possibility for direct observation (e.g. vortex studies) in a novel quantum system. We extended these links to include strongly correlated systems. Examples of phenomena we studied in this context include the superfluid to Mott insulator transition and the fractional quantum Hall effect.

The objective of this research program was to carry out a number of specific investigations to explore the physics of strongly correlated quantum systems in the dilute gas realm. The necessary prerequisite theoretical foundation was

Task Description:	developed. The renormalized quantum field theory for treating resonance superfluid systems has been a focus topic where strong agreement between theory and experiment has been established. The objective here was to extend this general framework to incorporate the areas of quantum Hall physics and quantum phase transitions which are important and of broad interest in condensed matter physics. The specific problems addressed were the treatment of resonant scattering in optical lattices (in which atoms move in an optical potential pattern formed by intersecting laser beams), in highly rotating quantum gases, and in strongly coupled fermion superfluids. This research program will have significant implications for understanding a broad spectrum of physical phenomena. Superfluidity is a subject in physics of fundamental importance, with ties to macroscopic quantum phenomena, to superconductivity, and to optical lasers. The most important work in the field of Bose-Einstein condensation in dilute gases is yet to come, because the recently demonstrated ability to provide quantum control of the interactions between the atoms allows potentially a new area of quantum state engineering in a quantum superfluid. The first topics to pursue are the strongly interacting systems relevant to condensed matter physics. These considerations are of such fundamental nature that it seems likely that technologies will be spawned of industrial importance; in metrology, in remote sensing, in quantum logic, and in mesoscale physics, to name a few.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cold quantum atomic systems have potential impact in a wide range of precision metrology and inteferometric measurement systems with many important applications to both earth and space sciences. In much the same way as the laser has impacted a wide range of technological fields, atomic condensates, which form a coherent matter equivalent to the coherent light, may potentially be useful in a broad range of applied phenomena. This includes both contribution to the fundamental understanding of technologically important systems (such as high Tc superconducting materials) by replicating the physical mechanisms in a highly controllable quantum system, and the ability to control precision measurements through the development of atomic lasers, atomic interferometry, and high-precision spectroscopy, in an unprecedented manner.
Task Progress:	Our research program concentrated on two primary aspects of strongly correlated physics in dilute Bose and Fermi gases. The first of these topics explored the effect of rotation as a virtual magnetic field for strongly interacting bosons in optical lattices. We had a number of successful results on this topic and predicted some remarkable theoretical phenomena which we expect to emerge in these systems. We demonstrated that rotation can introduce complexity and rich physics to the quantum phase transitions that occur for bosonic atoms in optical lattices. We showed that the ground state of the many-body system undergoes symmetry transitions at specific values of the rotation angular frequency. In effect, this corresponds to the introduction of vortices into the system of atoms in optical lattices. This system is quite different to the regular dilute Bose gas where vortices have been previously studied experimentally, as one can vary the effect of particle interactions by increasing or decreasing the lattice intensity. Our work demonstrated the possibility of exploring the interesting physics of the fractional quantum Hall effect for atoms in optical lattices where the physical system is quite different to the two-dimensional electron gas where the effect was first observed and studied. The second major topic concerns strongly correlated physics in fermion gases. This was a joint theory/experiment research effort which demonstrated a number of quantitative aspects of the study of fermions in the crossover between Bose-Einstein condensation and BCS superfluidity (the physics of superconductors). In spite of decades of work, the theoretical study of the BEC/BCS crossover remains an open problem. It provides an exceptional challenge to theorists, primarily due to the lack of a small parameter in which a perturbation expansion can be made. For this reason, theoretical studies to date have been primarily based on two variational approaches. The first is founded on the early work of Eagles and Leggett, taking a BCS wave function for the ground state form. The BCS ground state has been qualitatively successful in describing the crossover. The second main approach is based on quantum Monte Carlo calculations, also variational in spirit due to the need to fix the nodes in the many body wave function in order to solve the well known "sign" problem. Our work on this problem has been based on the BCS wave function approach. On both extreme limits of the crossover, where BCS theory applies and where BEC theory applies, the interactions are weak and we expect a mean-field description to be an accurate formulation. We recently showed that the BCS ground state wave function approach describes correctly the emergence of the Bose-Einstein condensate in the crossover as anticipated, but there is a major flaw in the theory in that the leading order term describing the interactions between the particles is not right. This implies an incorrect equation of state for the system in the bosonic limit. We believe this to be quite unsatisfactory since we know the mean-field description to be good for that regime. The incorrect equation of state means that all the thermodynamic quantities which emerge for the dilute Bose gas will be incorrect. Thus we posed an important question; can one derive a mean-field approach to the crossover, which at least reproduces correctly the weakly-interacting limits of the bosonic and fermionic gases? Our research investigated one of the most important quantities; the released energy in the strongly interacting crossover region. We showed that the most elementary theory which is able to encapsulate many agreeable qualitative features of crossover obtains the release energy incorrectly. This is significant because it means a more sophisticated picture is required and there exist interesting quantum correlations in the ground state of the fundamental crossover problem. The bigger picture here is that this work is starting to demonstrate the possibility for realizing many fundamental phenomena which lie at the forefront of condensed matter physics in the atomic gas environment. The need to obtain extremely clean systems working at very low temperatures so that thermal effects do not destroy the strongly correlated states one is investigating provides the motivation for exploring this kind of topic within the NASA program where the low gravity, low noise, and long integration time environment, is a perfect framework for these systems. The postdoctoral researcher supported on this grant (Simon Gardiner) was appointed in a tenure track position at the University of Durham in England. The grant contributed to the support of graduate students at the University of Colorado. The results of the research were disseminated at a number of colloquia, invited talks and contributed talks in major conferences. We were active in a number of outreach activities. These primarily consisted of public lectures (for example the

	Principal Investigator held a public discussion with about 50 members of the general public at the Aspen Center of Physics in 2005), incorporating material from this research project in the large (600+) undergraduate Freshman physics classes at the University of Colorado, and in the training and development of graduate and postdoctoral researchers. We also contributed to the JILA institute programs; which support other outreach activities, including the University of Colorado Wizards series, physics education research initiatives, and science presentations to the general public on a diverse set of topics ranging from Black Holes to Bose-Einstein condensation.
Bibliography Type:	Description: (Last Updated: 02/19/2008)
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